# Executive Summary for Government Stakeholders: Leveraging Machine Learning and Explainable AI for Workforce Retention and Fiscal Optimization

**Principal Data Scientist & Policy Advisor**
Advanced Analytics & Public Policy Directorate

June 7, 2026

**Abstract**

This executive summary translates complex analytical pipelines into actionable public-sector policy. Based on a comprehensive review of three interconnected data science workflows—Exploratory Data Analysis (EDA), Feature Engineering, and Multi-Model Classifier Training—this report establishes a highly rigorous framework to predict and mitigate voluntary employee attrition. Utilizing a dataset of $1,470$ employees, we identify critical operational bottlenecks, such as overtime and early-career business travel strains, and demonstrate how advanced Machine Learning (ML) classifiers (Logistic Regression, Random Forest, Gradient Boosting, and XGBoost) can predict turnover with high fidelity. Crucially, we leverage Explainable AI (XAI) frameworks—specifically SHAP (SHapley Additive exPlanations), DiCE (Diverse Counterfactual Explanations), and Partial Dependence Plots (PDP)—to provide policy makers with complete transparency. This ensures that algorithmic recommendations are legally compliant, auditable, and ethically aligned with public sector governance.

## 1 Introduction & Policy Context

For government stakeholders and public sector administrators, human capital is the single largest driver of operational capability and fiscal expenditure. Voluntary employee attrition imposes severe direct costs—including recruitment, onboarding, and training—and substantial indirect costs, such as the loss of institutional memory, degraded public service delivery, and diminished team morale.

Historically, public sector human resource (HR) departments have relied on retrospective exit surveys to understand turnover. This document introduces a proactive, predictive paradigm. By analyzing the entire analytical lifecycle across three technical notebooks, we present an end-to-end framework that allows government agencies to:

1. **Anticipate Attrition**: Identify high-risk roles and individuals before resignation occurs.
2. **Optimize HR Budgets**: Direct retention resources and salary adjustments to where they yield the highest return on investment (ROI).
3. **Support Labor Equity**: Establish evidence-based policies regarding overtime and business travel demands.

## 2 Methodological Rigor: The Machine Learning Pipeline

To ensure the integrity of any policy based on machine learning, the underlying data pipeline must be technically sound, reproducible, and robust against common biases. The pipeline consists of three sequential phases.

## 2.1 Phase 1: Exploratory Data Analysis (EDA)

Initial profiling of the $1,470$ workforce records revealed a baseline attrition rate of $16.12\%$ ($237$ individuals departed; $1,233$ remained active). This represents a highly imbalanced class distribution, a critical factor that informed our downstream modeling choices.

This baseline analysis establishes the primary characteristics of the active and departed cohorts, allowing us to identify key socio-demographic and workplace stressors that correlate with voluntary turnover. By checking data integrity, examining feature distributions, and verifying that no missing values exist across the feature space, we established a robust foundation for subsequent feature engineering and modeling stages.

## 2.2 Phase 2: Advanced Feature Engineering & Imbalance Mitigation

To prepare the cleaned dataset for classification, a structured, reproducible feature engineering pipeline was designed:

- **Numerical Processing**: Continuous features (e.g., `Age`, `MonthlyIncome`, `TotalWorkingYears`) were scaled using the `StandardScaler` to standardize the mean to $0$ and variance to $1$. This ensures that distance-based and gradient-descent algorithms are not dominated by large numerical ranges.

- **Categorical Encoding**: Discrete variables (e.g., `BusinessTravel`, `Department`, `JobRole`, `OverTime`) were converted to sparse representations using `OneHotEncoder(drop='first')` to eliminate collinearity and maintain model tractability.

- **Class Imbalance Resolution (SMOTE)**: Training classifiers on the raw $84\%$ to $16\%$ split would cause the models to default to a "No Attrition" prediction. To resolve this, we implemented the Synthetic Minority Over-sampling Technique (SMOTE) strictly on the training set:
$$(X_{\text{train\_final}}, y_{\text{train\_final}}) = \text{SMOTE}(X_{\text{train\_preprocessed}}, y_{\text{train}}) \tag{1}$$
This balanced the target classes by generating realistic synthetic records for the minority class, forcing the models to learn the distinct signatures of voluntary departure.

## 2.3 Phase 3: Multi-Model Evaluation

Four distinct machine learning architectures were trained and validated on a $20\%$ hold-out test set to identify the optimal model. Each model represents a different mathematical approach to classification:

1. **Logistic Regression**: Serves as a linear baseline, optimized using L2 regularization.

2. **Random Forest Classifier**: An ensemble of bootstrap-aggregated (bagged) decision trees, highly resilient to overfitting.

3. **Gradient Boosting Classifier**: A sequential boosting model that minimizes loss by fitting trees to the residuals of prior iterations.

4. **XGBoost Classifier**: An optimized, regularized gradient boosting implementation designed for speed and state-of-the-art predictive performance.

To ensure balanced performance, models were evaluated across multiple metrics, including Precision (to minimize false positives), Recall (to minimize missed attrition risks), F1-Score (the harmonic mean), and the Area Under the Receiver Operating Characteristic Curve (ROC AUC), which measures overall discriminative power.

# 3 Explainable AI (XAI) & Policy Transparency

For government entities, deploying a "black box" algorithm is politically and legally non-viable. Algorithmic decisions must be auditable, explainable, and compliant with civil rights and labor standards. We integrated three state-of-the-art XAI frameworks to inspect and validate our models.

## 3.1 SHAP (SHapley Additive exPlanations)

Based on cooperative game theory, SHAP calculates the exact marginal contribution of each feature to individual predictions. This provides a clear global and local hierarchy of attrition drivers, as summarized in Table 1.

Table 1: Global Feature Importance Hierarchy (SHAP Insights)

| Feature Rank | Feature Name | Nature of Algorithmic Influence |
|---|---|---|
| 1 | `OverTime_Yes` | Strong positive impact; increases attrition probability. |
| 2 | `MonthlyIncome` | Strong negative impact; lower income drives attrition. |
| 3 | `StockOptionLevel` | Strong negative impact; zero/low options drive attrition. |
| 4 | `YearsWithCurrManager` | Negative impact; short manager tenure increases risk. |
| 5 | `BusinessTravel_Frequently` | Positive impact; high travel demands drive attrition. |
| 6 | `MaritalStatus_Single` | Positive impact; single employees show higher mobility. |

## 3.2 DiCE (Diverse Counterfactual Explanations)

While SHAP explains *why* a prediction was made, DiCE answer the critical policy question: *"What must change to reverse this outcome?"*

DiCE generates diverse, actionable counterfactual scenarios for individual high-risk employees. For example, for a critical specialist predicted to leave with $82\%$ confidence, DiCE demonstrates that:

$$\text{Prediction}(\text{Exit}) \rightarrow \text{Prediction}(\text{Retain}) \quad \text{if} \quad \begin{cases} \texttt{OverTime} & \text{changed to No} \\ \texttt{MonthlyIncome} & \text{increased by } \$450 \\ \texttt{StockOptionLevel} & \text{increased to 1} \end{cases} \tag{2}$$

This allows HR managers to design highly personalized, legally defensible, and minimally disruptive retention plans for critical personnel.

## 3.3 Partial Dependence Plots (PDP)

PDPs analyze the isolated relationship between key continuous variables and the predicted probability of attrition, yielding two key insights:

- **The Compensation Saturation Curve**: Attrition risk drops steeply as `MonthlyIncome` increases from $\$1,500$ to $\$6,500$. Beyond $\$10,000$, the curve flattens completely, indicating that marginal salary increases above this threshold yield diminishing returns for retention.
- **The Commute Strain**: The predicted probability of attrition rises linearly with `DistanceFromHome`, confirming that geographical commute strain functions as a steady catalyst for departure.

# 4 Strategic Recommendations for Government Stakeholders

Based on this comprehensive analysis, we outline four strategic recommendations to modernize workforce retention policies:

## 4.1 1. Budgetary Realignment: Target the Lower-Income Attrition Threshold

- **Policy**: Implement targeted compensation adjustments for positions earning below $\$3,500$ per month, particularly at `JobLevel 1` and `JobLevel 2`.
- **Fiscal Justification**: Given that the cost of replacing an employee is approximately 1.5 times their annual salary, increasing the monthly salary of high-risk junior staff by $\$300$ to $\$500$ is far more cost-effective than absorbing the direct and indirect expenses of a turnover event.

## 4.2 2. Labor Policy Reform: Restrict Overtime Exploitation

- **Policy**: Establish a strict regulatory cap on consecutive overtime hours and introduce mandatory Compensatory Time-Off (TOIL) frameworks.
- **Fiscal Justification**: Overtime is the single largest predictor of attrition in the dataset (nearly tripling the rate to $30.5$%). Auditing departments with chronic overtime and expanding headcounts is a financially sound long-term strategy that reduces expensive burnout cycles.

## 4.3 3. Operational Mobility: Age-Segmented Travel Protections

- **Policy**: Restrict travel mandates for early-career cohorts (under 30 years old), utilizing virtual-first protocols for junior roles. Allocate travel responsibilities to senior staff (aged 41–60) who exhibit higher travel resilience ($10.5$% travel-related attrition).
- **Fiscal Justification**: Eliminating travel for vulnerable junior demographics directly protects the organization's entry-level talent pipeline, stabilizing long-term workforce planning.

## 4.4 4. Establish transparent, Explainable AI Governance

- **Policy**: Standardize the use of SHAP, DiCE, and PDP within public sector HR analytics to ensure all machine learning tools remain transparent and auditable.
- **Fiscal Justification**: Transparent algorithms protect government agencies from legal liabilities and compliance penalties, while ensuring that workforce interventions are demonstrably fair and evidence-based.

# 5 Conclusion

By integrating Exploratory Data Analysis, Feature Engineering, and Multi-Model Classifier Training with Explainable AI frameworks, this report provides government stakeholders with an empirically validated roadmap to optimize retention. Moving from a reactive posture to a proactive, transparent, and data-driven workforce strategy will safeguard public resources, preserve critical institutional knowledge, and ensure the stable and efficient delivery of public services.